RC Filter Frequency Response (Bode Plot)

1. Definition

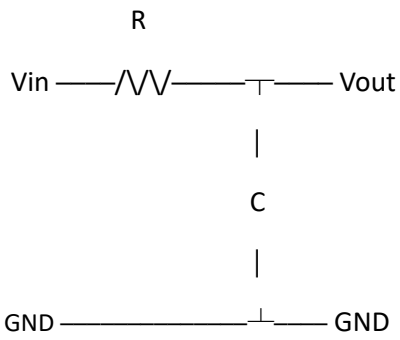
An RC filter is an electronic circuit consisting of a resistor (R) and a capacitor (C) that allows certain frequencies to pass while attenuating others. Its frequency response shows how the output voltage changes with input frequency.

2. Types of RC Filters

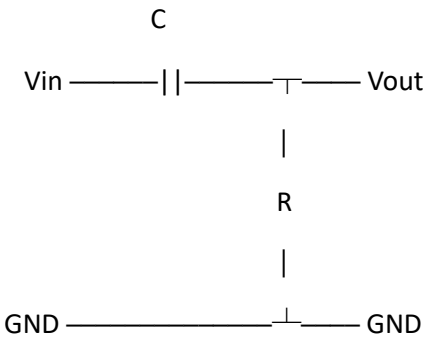
- Low-pass filter: Passes low frequencies and attenuates high frequencies.
- High-pass filter: Passes high frequencies and attenuates low frequencies.

3. Circuit Diagram

Low-pass RC filter



High-pass RC filter



4. Important Formulae

Cut-off frequency:

$f_c = 1/(2\pi RC)$

Low-pass filter gain:

$|H(f)| = 1/\sqrt{(1 + (f/fc)^2)}$

High-pass filter gain:

$|H(f)| = (f/fc)/\sqrt{(1 + (f/fc)^2)}$

Gain in decibels:

Gain (dB) = $20 \log_{10}(Vout/Vin)$

At the cut-off frequency:

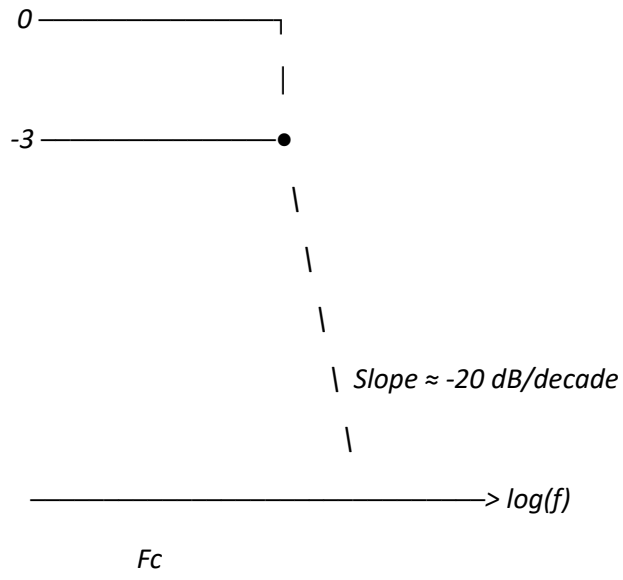
Vout = 0.707 Vin

Gain = −3 dB

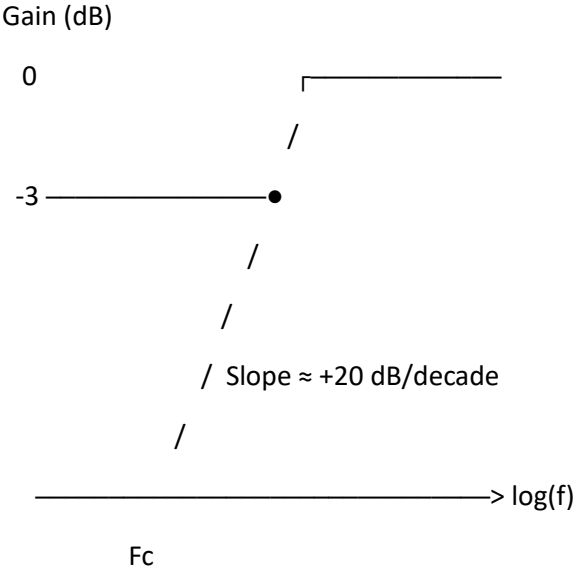
5. Bode Plot

A Bode plot represents the gain in decibels against frequency on a logarithmic scale.

Low-pass filter response



High-pass filter response



6. Observation Table

The following are illustrative values for a low-pass RC filter with fc = 1 kHz.

Frequency (Hz)	Gain (Vout/Vin)	Gain (dB)
100	0.995	−0.04
500	0.894	−0.97
1000	0.707	−3.01
2000	0.447	−6.99
5000	0.196	−14.15
10000	0.100	−20.04

7. Procedure

1. Connect the resistor and capacitor in the required filter configuration.
2. Apply a sinusoidal input signal of constant amplitude.
3. Vary the input frequency over a suitable range.
4. Measure the input and output voltages at each frequency.
5. Calculate the gain using Vout/Vin.
6. Convert the gain into decibels.

7. Plot gain (dB) against log frequency and identify the cut-off frequency.

8. Result

The frequency response of an RC filter is obtained by plotting gain against logarithmic frequency. The cut-off frequency occurs at approximately −3 dB, where the output voltage is 70.7% of the input voltage.